SAHIL POTALE

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SUMMARY

Final-year Information Technology graduate (BE, 2026) with hands-on project experience in machine learning, NLP, and full-stack development. Built ML models for real-world prediction tasks and an AI-driven system that maps requirements documents to cloud architecture. Actively upskilling in Generative AI and MLOps. Seeking a Trainee/Associate role in ML Engineering, Python Development, or Data Analysis.

TECHNICAL SKILLS

Languages: Python, JavaScript

ML / Data: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Transfer Learning (EfficientNetB0, VGG16, InceptionV3), Gradio, NLP, Feature Engineering, Data Preprocessing

Generative AI: LangChain (LCEL), RAG, HuggingFace Embeddings, ChromaDB, Groq, Streamlit

Web / Frontend: React.js, Vite, Flask, HTML5, CSS3

Integrations: Sanity CMS, WhatsApp API, Google Maps API

Tools: Git, GitHub, AWS, Azure (cloud service mapping)

PROJECTS

LungCare – Lung CT Image Classification *Jun 2025 – Aug 2025*

- Developed a deep learning system to classify lung CT images into 4 classes: Adenocarcinoma, Large Cell Carcinoma, Normal, and Squamous Cell Carcinoma
- Implemented transfer learning with EfficientNetB0, achieving 95.87% classification accuracy, outperforming VGG16 and InceptionV3 in benchmarking
- Built an interactive Gradio web app for CT image upload with real-time model prediction and confidence scores

Chat with Your Documents – RAG-based PDF Q&A (Live Demo) *Sep 2026 – Oct 2026*

- Built a Retrieval-Augmented Generation (RAG) app that answers natural-language questions about any uploaded PDF, grounded in the document's actual content
- Implemented document chunking and free local HuggingFace embeddings (all-MiniLM-L6-v2), with ChromaDB as the vector store for similarity search
- Orchestrated retrieval and generation using LangChain (LCEL) with a Groq-hosted LLM, built the UI with Streamlit, and deployed it live with per-upload isolated vector stores

Designing with Cloud: SRS to System Architecture *Jul 2025 – Mar 2026*

- Built an AI-powered system that parses Software Requirements Specifications (SRS) using NLP and recommends matching AWS/Azure cloud services
- Engineered requirement-extraction logic to classify functional vs. non-functional requirements and map each to a suitable cloud service
- Designed automated cloud architecture generation with cost estimation and Infrastructure-as-Code export

Melophile — Music Academy Website (Live Client Project) *Aug 2026 – Present*

- Built and deployed a production website for a real client (Melophile Music Academy) using React and Vite, live at a custom domain
- Integrated Sanity CMS for dynamic blog content management, a WhatsApp-based student inquiry system, and Google Maps for location

EDUCATION

Vidyavardhini's College of Engineering & Technology — BE, Information Technology *Sep 2023 – May 2026*
GPA: 7.08

Pravin Patil College of Diploma Engineering & Technology — Diploma, Computer Engineering *2020 – 2023*
Percentage: 81.66%

CERTIFICATIONS & CONTINUOUS LEARNING

- Complete Generative AI Course with LangChain and HuggingFace — Udemy (In Progress, 50%)
- Complete MLOps Bootcamp with 10+ End-to-End ML Projects — Udemy (In Progress, 35%)